

MARL-based Pricing Strategy via Mutual Attention for MoD Systems with Ridesharing and Repositioning

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Abstract—The pricing strategy plays a vital role in increasing the revenue of mobility on-demand (MoD) systems by balancing supply-demand relationship across urban zones. MoD systems often incorporate both ridesharing and vehicle repositioning to maintain this balance, ultimately improving revenue and order completion rate. However, many existing pricing strategies overlook the effects when the price differences across zones meet ridesharing and repositioning, leading to supply-demand mismatch and reduced revenue. To tackle this problem, this paper presents a multi-agent reinforcement learning (MARL) based pricing strategy, named MAP, which employs a mutual attention mechanism to effectively account for price differences with ridesharing and repositioning. We transform the pricing problem as a MARL model that aiming to maximizing total revenue. These agents are tasked with making about order fare with ridesharing, vehicle income with repositioning for each zone. Pricing strategy decision-making is guided by mutual information theory and further enhanced by an attention mechanism to estimate how pricing variations impact adjacent zones. Simulation studies utilizing real-world datasets are performed to illustrate the advantages of MAP compared to existing benchmarks.

Index Terms—Mobility on-Demand, Pricing, Supply-demand Balance.

I. INTRODUCTION

Evolutionary mobility on-demand (MoD) systems, e.g., Uber and DiDi, have successfully bridges order demand and vehicle supply, creating two-sided ride-hailing markets that are both socially and commercially valuable [1], [2]. With MoD systems, the passengers request vehicle services by paying a fare as the vehicle's income and the platform's revenue. The primary goal of MoD platforms is to optimize total revenue by fine-tuning pricing strategies for both order fares and vehicle earnings [3]–[5].

The overall revenue of a MoD system depends on the order fare, vehicle income, and the number of completed orders. Simply increasing the order fare or decreasing the vehicle income does not necessarily lead to higher overall revenue. Since customer and driver behaviors are sensitive to pricing [3], very high order fares can cause customers to cancel orders, leading to fewer orders received by the

platform. Conversely, insufficient vehicle incomes may discourage vehicles from working, thereby decreasing the number of completed orders. In practice, both overly high fares for orders and inadequate incomes for vehicles leading to supply-demand imbalance, as well as revenue reduction. It is well-recognized that developing an effective pricing strategy that sustains supply-demand balance is essential to maximizing revenue in MoD platforms [6].

Supply-demand distribution often experiences imbalances at both spatial and temporal levels. For instance, during evening peak hours, there is often a concentration of orders in commercial centers. Most MoD systems divide a city into multiple smaller zones, creating zone-specific pricing strategies based on the unique supply-demand relationship [7], [8]. In zones facing supply shortages, increasing order fares can help prioritize urgent requests while discouraging less critical ones, thereby alleviating the undersupply issue. Conversely, in areas with excess supply, the platform should restrict the number of active vehicles to minimize unnecessary operational costs. Surge pricing is a widely used method that raises order fares during peak hours in undersupply areas based on empirical values [9], [10]. With the advancement of machine learning, multi-agent reinforcement learning (MARL) has become a popular tool for accurately determining optimal pricing strategies [10]. In MARL, each zone, characterized by a dynamic supply-demand relationship, is treated as an agent. These agents calculate the expected revenue under various pricing strategies, allowing for more precise and dynamic pricing adjustments. For example, Chen et al. [11] proposed integrating contextual bandit with temporal difference learning, where pricing strategies for orders are learned via a contextual bandit algorithm with the help of temporal difference learning.

However, it neglects that the price differences between zones can alter the supply-demand distribution. A zone that initially achieves the desired vehicle supply with its pricing strategy might lose vehicles to another zone offering more attractive incomes. Similarly, an high-demand zone may attract vehicles from other zones. Especially with the employment of the order ridesharing [12] and vehicle repositioning [13], this effect is further amplified. Ridesharing enables a vehicle to

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platform as an ordinary or a ridesharing order under pricing strategy $\mathbf{a}_t$. They are derived from Uber data to model the behavior of orders and vehicles in response to pricing, as detailed in Sec. IV.

In this context, we delineate the order fare strategy based on the pricing frameworks employed by Uber [16] and Lyft [17], as outlined below:

Definition 1 Order Fare Strategy

After the passenger are dropped off, they pay a fare that comprises a **base fare**, serving as the income for the assigned vehicle, along with an additional **service fare** for the MoD platform. This fare structure is determined by the pricing strategy applied to both ordinary and ridesharing orders.

- The base fare refers to a fixed price based on the travel distance. Let A represent the minimum fee per order, also referred to as the initial fare. Meanwhile, B denotes the additional fee for varying travel distances, while C signifies the discount rate applied to ridesharing. The order base fare is given by

$$e_o = \begin{cases} A + B \times d_o, & f_o = 1, \\ A + B \times C \times d_o, & f_o = 0. \end{cases} \quad (1)$$

- The service fare is collected by the MoD platform for offering the ride-hailing service. If the order is received as an ordinary order in zone p_o in time slot $t \in 1, K+1, 2K+1, \dots$, the service fare paid to the platform is defined by a rate, specifically, $\lambda_t^1(p_o)$, as

$$e_o^b = (\lambda_t^1(p_o) - 1) \times p_o, \quad (2)$$

Otherwise, the service fare paid to the platform as a ridesharing order is

$$e_o^r = (\lambda_t^2(p_o) - 1) \times p_o. \quad (3)$$

Here, $\lambda_t^2(p_o)$ indicates the discount of the ridesharing order compared to ordinary order as the passengers share a vehicle with others.

It should be noticed that the passengers prefer to delay or even cancel the order when suffer from high order fares.

B. Vehicles

The vehicles are defined as follows: Let $\mathbf{V}_t$ denote the potential vehicles in time slot t . Also, we use a 6-dimensional feature vector $\langle f_{v,t}, c_v, p_{v,t}, \bar{p}_{v,t}, \zeta_v(\mathbf{a}_t), \zeta'_v(\mathbf{a}_t) \rangle$ to denote vehicle v in time slot t , where

- $f_{v,t}$: denotes the vehicle state, where $f_{v,t} = 0$ indicates the vehicle is serving orders and $f_{v,t} = 1$ indicates the vehicle is repositioning;
- c_v : is the passenger capacity;
- $p_{v,t} \in \mathbf{N}$: is the zone index of vehicle v in time slot t ;
- $\bar{p}_{v,t} \in \mathbf{N}$: is the target zone of its serving order or repositioning. When the vehicle serves multiple ridesharing orders, the target zone is determined by the planning path of the orders;

- $\zeta_v(\mathbf{a}_t)$: is the probability that the vehicle accepts to serve orders under pricing strategy $\mathbf{a}_t$;
- $\zeta'_v(\mathbf{a}_t, n)$: is the probability that the vehicle is willing to reposition to zone n

Here, $\zeta_v(\mathbf{a}_t)$ and $\zeta'_v(\mathbf{a}_t, n)$ are also two price-sensitive functions [15]. Let $\beta_t(o, v)$ denote the dispatching decisions in time slot t . They are made by matching the order-vehicle bipartite graph [18]. Here, the edge weight in the bipartite graph is assigned based on the order fare (Definition 1). The vehicle can serve at most one ordinary order at same time, i.e.,

$$\sum_{o \in \mathbf{O}_t(\xi_o, \xi'_o, \mathbf{a}_t)} \beta_t(o, v) f_o \leq 1, \quad (4)$$

While for the ridesharing orders, the vehicle should ensure the total passenger number is less than its person capacity, i.e.,

$$\sum_{o \in \mathbf{V}_t(\zeta_v, \zeta'_v, \mathbf{a}_t)} \beta_t(o, v) c_o (1 - f_o) \leq c_v. \quad (5)$$

We define the supply-demand gap for zone n in time slot t as the difference between the number of orders and vehicles in that zone, i.e., $|\mathbf{O}_t(n)| - |\mathbf{V}_t(n)|$. When this value is negative, the zone is in oversupply; otherwise, it is in undersupply. Building on this, we introduce the cross-zone supply-demand gap $\Delta_t(n, n')$ between zone n and zone n' , which is

$$\Delta_t(n, n') = (|\mathbf{O}_t(n)| - |\mathbf{V}_t(n)|) - (|\mathbf{O}_t(n')| - |\mathbf{V}_t(n')|). \quad (6)$$

A higher value of $\Delta_t(n, n')$ suggests that the supply capacity in target zone n' exceeds that of zone n . In such cases, a higher repositioning fee should be offered to encourage vehicles to reposition to the target zone in an incentive manner.

Typically, all potential vehicles in time slot t determine, with a specific probability, whether to be scheduled for order or to reposition according to the vehicle income strategy.

Definition 2 Vehicle Income Strategy

Generally, this base fare A of the order is paid to vehicle v as the vehicle income. With MoD system, a portion of this income, however, should be allocated to the platform as a dispatching **service fee**. The remaining part of A , along with the repositioning fee provided by the platform, makes up the total income of vehicle v . When the vehicle has no order to serve, it may be scheduled for repositioning, for which the platform pays an additional **repositioning fee**.

- The service fee paid by vehicle v is

$$e_v^b = (1 - f_{v,t}) \times \mu_t^1(p_{v,t}) \times p_o, \quad (7)$$

where $\mu_t^1(p_{v,t})$ is the ratio of the service fee.

- The repositioning fee serves motivate the vehicle to reposition from the current location to zone n in an incentive manner, which is

$$e_v^r(n) = f_{v,t} \times \mu_t^2(p_{v,t}) \times \bar{\Delta}_t(p_{v,t}, n) \times Bd(p_{v,t}, n), \quad (8)$$

where $\mu_t^2(p_{v,t})$ is the incentive rate. To prevent the excessive attraction of specific zones, we normalize the cross-zone supply-demand gap $\Delta_t(p_{v,t}, n)$ to $\bar{\Delta}_t(p_{v,t}, n)$.

C. Pricing

The primary design behind the pricing strategy $\mathbf{a}_t$ is to regulate the supply-demand distributions towards supply-demand balance to maximize the total revenue. The pricing strategy in time slot t for all zones, $\mathbf{a}_t$, can be denoted by a 4-dimensional vector $\{\lambda_t^1, \lambda_t^2, \mu_t^1, \mu_t^2\}$. Note that the pricing strategy only update in time slot $t \in \{1, K+1, 2K+1, \dots\}$.

With a given pricing strategy, we can obtain the actually received orders, denoted by $\mathbf{O}_t(\xi_o, \xi'_o, \mathbf{a}_t)$ and served vehicle $\mathbf{V}_t(\zeta_v, \zeta'_v, \mathbf{a}_t)$ information.

The expected overall revenue of the platform in time slot t is composed of the income for providing services and the expense for encouraging repositioning. The expected income is further divided into the service fare of order o and the service fee of vehicle v , i.e.,

$$u_{t,1}(v, o) = \beta_t(o, v)[f_o e_o^b(1 - f_o)e_o^r + e_v^b]. \quad (9)$$

Normally, increasing the λ_t^1 , λ_t^2 and μ_t^1 also increase the earned revenue of platform per order.

The expected expense for encouraging vehicle v reposition to zone n is

$$u_{t,2}(v, n) = [1 - \sum_{o \in \mathbf{O}_t(\xi_o, \xi'_o, \mathbf{a}_t)} \beta_t(o, v)]e_v^r(n). \quad (10)$$

Therefore, the expected overall revenue can be calculated as follows

$$u_t = \sum_{v \in \mathbf{V}_t(\zeta_v, \zeta'_v, \mathbf{a}_t)} \left\{ \sum_{o \in \mathbf{O}_t(\xi_o, \xi'_o, \mathbf{a}_t)} u_{t,1}(v, o) - \sum_{n=1}^N u_{t,2}(v, n) \right\}. \quad (11)$$

Here, we use $u_t(n)$ to represent the reward earned by the orders and vehicles in zone n . Increasing μ_t^2 means the platform should expend the earned revenue of platform per repositioning.

The pricing with ridesharing and repositioning for MoD systems can be formulated as a revenue-maximum optimization problem as follows:

$$\mathbf{P1} : \quad \max_{\mathbf{a}_t} \sum_{t=1}^T u_t \quad (12)$$

$$s.t. \quad (4), (5) \quad (13)$$

It is difficult to solve the above optimization problems using existing methods due to the following reasons. First, let $|\lambda|$ and $|\mu|$ denote the feasible solution spaces of pricing strategy for orders and vehicles. The pricing strategies for orders involving ridesharing and for vehicles with repositioning are greatly coupled, resulting in considerable complexity of $O(N|\lambda|^2|\mu|^2T)$. Especially with taking dispatching decisions into account, the complexity of solving **P1** will reach up to $O(N|\lambda|^2|\mu|^2T|\mathbf{V}_t|^{3WT})$.

Secondly, since the potential orders and vehicles arrive in an online manner, i.e., the platform has limited information on future orders and vehicles. Consequently, formulating a pricing

strategy for subsequent periods complicates the derivation of an optimal solution when addressing the problem online.

Hence, we transform **P1** into a multi-agent extension of partially observable Markov decision process (POMDP), i.e., $\langle \mathcal{S}_w, \mathcal{Z}_t, \mathcal{A}_t, \mathcal{R}_t \rangle$:

- **Environment State $\mathcal{S}_t$:** We represent the geographic characteristics of zone n by a 7-dimensional vector $\mathbf{g}_n$, which includes factors such as time, day or night, holiday status, weekday, temperature, wind, and weather. These features have a significant impact on both the potential supply-demand distribution [19]. Additionally, as the orders and vehicles have multiple attributes, using all detailed information as environmental states is not practical.

Instead, we simplify the representation by quantifying the number of vehicles and orders in each zone in time slot t . The state $\mathbf{s}_t$ is then represented as

$$\mathbf{s}_t = \left\{ |\mathbf{O}_t(1)|, \dots, |\mathbf{O}_{t+K-1}(N)|, |\mathbf{V}_t(1)|, \dots, |\mathbf{V}_{t+K-1}(N)|, \mathbf{g}_1, \dots, \mathbf{g}_N \right\}. \quad (14)$$

- **Observation $\mathcal{Z}_t$:** The impact of pricing strategies across different zones considerably reduces as the distance between those zones grows. Let $\mathbf{z}_t^n \in \mathcal{Z}_t$ denote the partial observation of zone n , which is

$$\mathbf{z}_t^n = \{ |\mathbf{O}_t(n)|, \dots, |\mathbf{O}_{t+K-1}(n)|, \quad (15)$$

$$|\mathbf{V}_t(n)|, \dots, |\mathbf{V}_{t+K-1}(n)|, \mathbf{g}_n \}. \quad (16)$$

Note that the complete environment state is the composition of all users' observed states.

- **Pricing Action $\mathcal{A}_t$:** The overall pricing action is $\mathbf{a}_t \in \mathcal{A}_t$. For zone n , we expand it to an action for each agent, i.e.,

$$\mathbf{a}_t^n = \{ \lambda_t^1(n), \lambda_t^2(n), \mu_t^1(n), \mu_t^2(n) \}. \quad (17)$$

- **Reward $\mathcal{R}_t$:** The reward is a function of observation and action, which measures the effect of the action. Commonly, the goal of dynamic pricing is to maximize revenue r_t^n according to Eq. (11)

$$r_t^n = \sum_{t'=t}^{t+K-1} u_{t'}(n). \quad (18)$$

III. MAP STRATEGY

In this section, an MARL approach, called MAP, is leveraged here, where each zone acts as an agent to learn its own pricing strategy with ridesharing and repositioning.

It can be seen from Fig. 2, each agent makes its pricing strategy, i.e., $\mathbf{a}_t^n$, based on observation $\mathbf{z}_t^n$. These agents obtain their local Q-values $Q_t^n(\mathbf{z}_t^n, \mathbf{a}_t^n, \theta_n)$, i.e.,

$$Q_t^n(\mathbf{z}_t^n, \mathbf{a}_t^n, \theta) = \mathbb{E}[r_t^n + \sum_{i=1}^{\infty} \gamma_i r_{t+i}^n | \mathbf{z}_t^n, \mathbf{a}_t^n], \quad (19)$$

by two Multi-layer Perceptions (MLPs) whose parameters are denoted by θ according to its policy. The first MLP is constructed to learn the pricing strategy for ordinary order fares and vehicle incomes, i.e., $\lambda_t^1(n)$ and $\mu_t^1(n)$. These values

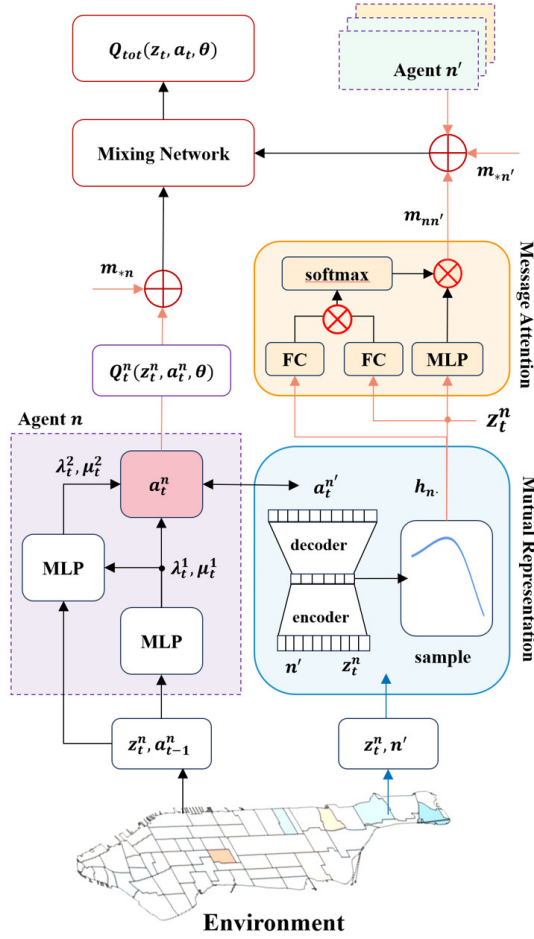


Fig. 2: The illustration of MAP strategy.

also serve as inputs to the second MLP, which determines the pricing strategy for ridesharing orders and vehicle repositioning, i.e., $\lambda_t^2(n)$ and $\mu_t^2(n)$. This approach enables the capture of the coupled pricing strategy with ridesharing and vehicle repositioning.

To account for the mutual effect induced by price differences among agents (zones), we estimate the mutual information between each agent to generate a **mutual representation** for communication with others. These generated messages are subsequently fed into the policies of other agents as incentive factors, which help update their local Q-values to $\bar{Q}_t^n(z_t^n, a_t^n, \theta)$. As the mutual effect diminishes with the increasing distance between zones, we introduce the **message attention** mechanism to effectively evaluate the relative importance of each message and adjust its actions accordingly.

With the incentive mutual messages, the local Q-values integrate with each other by a QMIX [20] network to maximize the global value

$$Q_{tot}(z_t, a_t, \theta) = \mathbb{E}[\sum_{n=1}^N \bar{Q}_t^n(z_t^n, a_t^n) | z_t^n, a_t^n]. \quad (20)$$

QMIX achieves an optimal solution by enhancing stability

via experience replay buffers and target networks, ensuring appropriate Q-learning updates. In MoD systems, the interaction between pricing strategies across zones can be modeled as a convex combination of local revenues to reach the optimal outcome. Furthermore, by incorporating the attention mechanism, the overall total Q-value becomes the weighted sum of the individual local Q-values. This relationship is supported by the findings in [21], which demonstrate the existence of specific weights that guarantee the optimality of Q_{tot} derived from the projection with the attention mechanism. Based on this, **P1** can be solved by a MARL method with a low complexity $O(|\lambda|^2|\mu|^2)$, where each zone makes its pricing strategy independently.

A. Mutual Representation

To capture the influence of actions taken by other agents, we introduce a mutual representation model [22], which quantifies the effect of price differences between zones. Specifically, this model maps the local observation z_t^n and another agent's zone index n' into a mutual representation $h_{nn'}$, which characterizes the influence of agent n on agent n' .

To emphasize spatial relations, the model encodes n' and z_t^n into a multivariate Gaussian distribution $\mathcal{N}(\mu_{n,n'}, \delta_{n,n'}^2)$ [23]. Sampling from this distribution generates a valid mutual representation $h_{nn'}$ using a variational distribution $q(h_{nn'} | n')$. This representation captures both bidirectional influence and spatial dependencies, thereby improving the adaptability of strategy formulation in multi-agent systems.

Let $H(\cdot)$ denote entropy. Since the selected pricing actions intuitively reflect the level of coordination among agents, the learned mutual representation should be sensitive to action selection. As depicted in Fig. 3, the mutual representation $h_{nn'}$ and the pricing action $a_t^{n'}$ jointly affect the uncertainty of platform's revenue, denoted as $H(h_{nn'}, a_t^{n'} | z_t^n, n')$. **Mutual Information (MI)** can be interpreted as the degree to which knowing information about $a_t^{n'}$ reduces uncertainty about $h_{nn'}$, i.e.,

$$I(h_{nn'}; a_t^{n'} | z_t^n, n') = H(h_{nn'} | z_t^n, n') - H(h_{nn'} | a_t^{n'}, z_t^n, n'). \quad (21)$$

When the correlation between $h_{nn'}$ and $a_t^{n'}$ is low, the entropy $H(h_{nn'} | z_t^n, n')$ approaches the conditional entropy $H(h_{nn'} | a_t^{n'}, z_t^n, n')$. Maximizing MI thus corresponds to minimizing the uncertainty about agent interactions given local information.

However, it is difficult to compute the conditional distribution directly. Hence, the decoder takes $h_{nn'}$ as input to learn $p(h_{nn'} | a_t^{n'}, z_t^n, n')$ for approximating the distribution $q(h_{nn'} | z_t^n, n')$. We advance a regularization by deriving a lower bound for MI [24], i.e.,

$$\begin{aligned} I(h_{nn'}, a_t^{n'} | z_t^n, n') & \geq \mathbb{E}[-D_{KL}(q(h_{nn'} | z_t^n, n') || p(h_{nn'} | a_t^{n'}, z_t^n, n'))] \\ & = \mathbb{E}[-D_{KL}(q(h_{nn'} | n') || p(h_{nn'} | a_t^{n'}, z_t^n, n'))], \end{aligned} \quad (22)$$

as z_t^n does not depend on the latent variable $h_{nn'}$ in the encoder. We utilize the lower bound to train the mutual

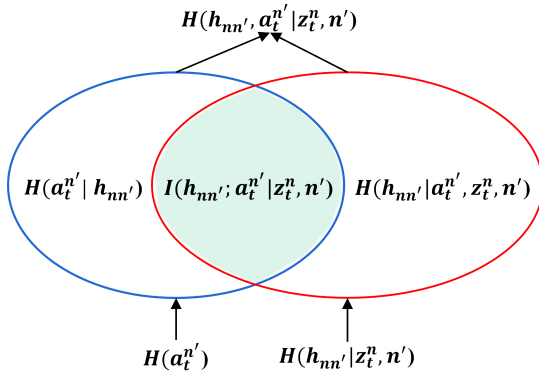


Fig. 3: Mutual information among local observation z_t^n , another agent's zone index n' and its action $a_t^{n'}$.

representation encoder parameterized by θ_m with a specific mutual loss, i.e.,

$$\mathcal{L}_m(\theta_m) = \sum_{n \neq n'} \mathbb{E}[D_{KL}(q(h_{nn'} | n')) || p(h_{nn'} | a_t^{n'}, z_t^n, n')]. \quad (23)$$

B. Message Attention

The effect of pricing strategies among agents is highly sensitive to spatial factors, as longer travel distances hinder repositioning. To reduce the weak impact of remote zones, an attention mechanism filters out less relevant pricing signals. Before sending mutual messages, agents apply a self-attention mechanism [25] to assign adaptive weights the outgoing messages, thereby improving the efficiency of message aggregation by emphasizing the most relevant interactions.

To model the unique spatial-temporal dependencies among agents, we use two distinct Fully Connected (FC) layers, each designed to process the mutual representation $h_{nn'}$ and the local observation z_t^n independently. They aim to extract relevant features by learning a query vector q_n and a key vector $k_{nn'}$. The query captures the contextual data of the local agent, whereas the key encodes the interactions between the agent and its neighboring counterpart. The weight parameter for the interaction between agents n and n' can be obtained by:

$$\omega_{nn'} = \text{softmax}(q_n^T k_{nn'}), \quad (24)$$

Specifically, we transform the mutual representation $h_{nn'}$ and local observation z_t^n into the final mutual message $v_{nn'}$. This message, aligned with the local Q-network dimension by a MLP, ensures effective integration of spatial and local information for decision-making. Therefore, the final attention-based message is

$$m_{nn'} = \omega_{nn'} v_{nn'}. \quad (25)$$

We utilize the final message to bias the other agent's Q-values as an incentive. The final Q-value of agent n is

$$\bar{Q}_t^n(z_t^n, \cdot, \theta) = Q_t^n(z_t^n, \cdot, \theta) + \sum_{n \neq n'} m_{nn'}, \quad (26)$$

Algorithm 1 MAP Strategy

Input: : State s_t , mutual representations z_t^n

Output: : Actions a_t^n , rewards r_t^n

```

1: for  $t = 1$  to  $T$  do
2:   for  $n = 1$  to  $N$  do
3:     Obtain local Q-value using Eq. (19);
4:     Generate mutual representation via mutual encoder;
5:     Aggregate mutual messages according to Eq. (25);
6:     Update local Q-value according to Eq. (26);
7:   end for
8:   Obtain global Q-value based on Eq. (20);
9:   Evaluate overall loss using Eq. (30).
10: end for

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where $\bar{Q}_t^n(z_t^n, \cdot, \theta)$ and $Q_t^n(z_t^n, \cdot, \theta)$ denote the corresponding Q-value vector for every action a_t^n .

To prevent assigning uniform weights to different agents, which typically results from the neural network's limited capacity to enforce sparsity, we introduce a sparsity regularization mechanism. This regularization aims to enhance the diversity of communication weights by minimizing the entropy of their categorical distribution, referred to as attention loss, i.e.,

$$\mathcal{L}_a(\theta_p) = \sum_{n=1, n \neq n'}^N \omega_{nn'} \log(\omega_{nn'}), \quad (27)$$

where θ_p is parameters of the attention-based pricing message generator. By cutting useless links whose weight is less than an empirical threshold, we can reduce the redundant information.

C. Loss Function

MAP strategy is implemented with the centralized training and decentralized execution (CTDE) paradigm. All parameters are updated by the following TD loss

$$\mathcal{L}_{TD}(\theta) = \mathbb{E}_{(z_t, a_t, r_t, z_{t+1}) \sim \mathcal{D}} \left[(y - Q_{tot}(z_t, a_t, \theta))^2 \right], \quad (28)$$

where

$$y = r + \max_{a'} Q_{tot}(z_{t+1}, a_{t+1}, \theta^-) \quad (29)$$

θ^- represents the parameters of the target network, which is updated periodically. Along with the previously discussed mutual loss and attention loss, the learning objective for MAP is defined as:

$$\mathcal{L}(\theta) = \mathcal{L}_{TD}(\theta) + \sum_{n=1}^N \sum_{i \in \{m, a\}} w_i \mathcal{L}_i(\theta_i), \quad (30)$$

where w_i is adjustable hyperparameters of mutual representation, and message attention, respectively. To prevent the lazy-agent problem and simplify the model architecture, the parameters of the message generator are also shared by agents. The detail procedure of MAP strategy can be found in Algorithm. 1.

IV. EXPERIMENT

Data Description: We conduct experiments with two city-scale Chengdu [26] and Manhattan [27] dataset. Chengdu dataset comprises over 6 million order records from Chengdu, spanning from November 1st to 30th, 2016. The map is uniformly divided into 16 square zones. Manhattan dataset consists of 1 million for-hire vehicle trip records within Manhattan, covering the period from January 1st to 30th, 2023. The provided road map is divided into 63 irregular zones.

We randomly generate the number of passengers of order $c_o \in \{1, 2\}$, and the person capacity of vehicle $c_v \in \{3, 4\}$. For the order fare price strategy, A, B, C are set to 10, 1.8, 0.5 [16]. The feasible action sets of pricing strategies [3] are $\lambda_t^i(n) \in \{0.7, 0.8, \dots, 1.3\}$, $i \in \{1, 2\}$, and $\mu_t^i(n) \in \{0, \dots, 0.4\}$, $i \in \{1, 2\}$.

We evaluate the MAP strategy over multiple days, averaging 24-hour results starting from midnight. The dataset is split into 70% for training, 10% for validation, and 20% for testing. Each day consists of 48 time slots of 30 minutes, with an episode length of 47 and a dispatching interval of 1 minute. The latent dimension of the mutual encoder is set to 8. In the message attention module, the MLP hidden layer size and attention dimension are 64 and 32, respectively. Mutual and attention loss weights are 0.001 and 0.01, respectively [22]. The actor and critic networks use a learning rate of 0.0005, and the reward discount γ is 0.99.

Behaviors of Orders and Vehicles: As the order is attracted by a low service fare, we use a price-sensitive exponential function [15] to denote the probability of a potential order converting into a received order, i.e.,

$$\xi_o(a_t) = \frac{1}{1 + \exp(0.67 \times \frac{\lambda_t^1(p_o) |\mathbf{V}_t(p_o)|}{|\mathbf{O}_t(p_o)|} - 1.67)}, \quad (31)$$

With the $\lambda_t^1(p_o)$ increasing, this probability gradually reduces. Meanwhile, the supply-demand distribution in the origin is taken into account, which relatively influences the pick-up time as well as the behavior of passengers. When the order fail to become an ordinary order, it has a probability of being ridesharing, i.e.,

$$\xi'_o(a_t) = \frac{1 - \xi_o(a_t)}{1 + \exp(0.67 \times \frac{\lambda_t^2(p_o) |\mathbf{V}_t(p_o)|}{|\mathbf{O}_t(p_o)|} - 1.67)}. \quad (32)$$

This is because that low service fares for ridesharing encourage the passengers to share a vehicle. Similarly, we define the probability that the vehicle is willing to work as

$$\zeta_v(a_t) = \frac{1}{1 + \exp(-0.67 \times \frac{\mu_t^1(p_{v,t}) |\mathbf{V}_t(p_{v,t})|}{|\mathbf{O}_t(p_{v,t})|} - 1)}, \quad (33)$$

as a high income impedes the drivers to work. Additionally, if the vehicle is not dispatched to a certain order, it has a probability to reposition, i.e.,

$$\zeta'_v(a_t, \cdot) = \frac{\zeta_v(a_t)}{1 + \exp(-\sum_{n=1}^N g(p_{v,t}, n) - \mu_t^2(p_{v,t}) + 1)}, \quad (34)$$

where

$$g(p_{v,t}, n) = \max\left\{\frac{\lambda_t^2(p_{v,t}) - \lambda_t^2(n)}{d(p_{v,t}, n)}, 0\right\}. \quad (35)$$

the price difference between two zones increases, the probability of the vehicle repositioning to other zones increases. The probability of vehicle repositioning to zone $n \neq p_{v,t}$ is

$$\zeta'_v(a_t, n) = \frac{g(p_{v,t}, n') \times \zeta'_v(a_t, \cdot)}{\sum_{n'=1}^N g(p_{v,t}, n')}. \quad (36)$$

In practice, the probabilities are related to many factors, while in this paper we pays more attention to the effect of price strategy. Also, these exponential function can be replaced by some behavior learning methods, such as [28].

Baselines: We compare the performance of the MAP strategy with the three benchmarks given below:

- **Current Origin Pricing Policy (COP) [29]:** An algorithm uses the current demand and supply across the entire geographic zone to guide the pricing strategy decision-making, while ignoring the effects of ridesharing and vehicle repositioning.
- **AP-RTRS [30]:** The demand pattern of AP-RTRS is decided at the zone level, aiming to maximize the weighted sum of order completion rate and minimizes the relocation cost. AP-RTRS formulates the problem as a mixed integer linear program, and solves it by a column generation approach. Note that it also assumes all the vehicles are willing to obey the repositioning decisions.
- **ROSDR [29]:** It aims to maximize the revenue and order response rate based on soft actor-critic methods, where the ridesharing is taken into account. It assumes all the vehicles are willing to obey the reposition decision, and a fixed discount for ridesharing, i.e., $\lambda_t^2 = 1$.

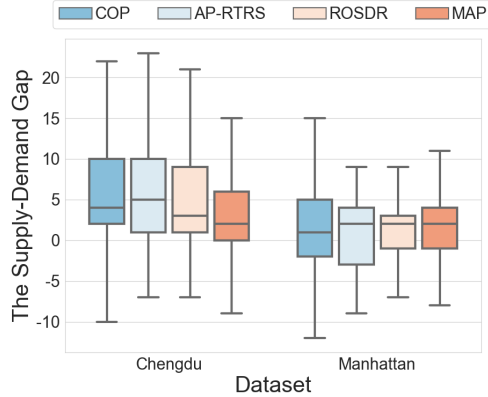
A. Overall Performance Comparison

Table. I shows the average daily revenue (ADR), the order acceptance rate (OAR), i.e., the percentage of the received orders in all the potential orders, and the order completion rate (OCR), i.e., the percentage of the served orders in all the potential orders in Chengdu and Manhattan datasets.

COP only makes the pricing strategy for ordinary orders and vehicles to serve orders based on the current supply-demand relationship, leading to the lowest ADR, OAR, and OCR. It is mainly caused by the limited capacity of COP to balance the dynamic supply-demand distribution without taking ridesharing and repositioning into account. With the ridesharing and repositioning, ROSDR and AP-RTRS have a certain elevated improvement in ADR and OAR due to their optimization objective and a tiny improvement in OCR. This is because the probability of vehicle repositioning is ignored, where no extra fee for repositioning also blocks the drivers from repositioning, which still leaves room for improvement. The MAP strategy reaches the highest performance as it introduces the pricing message between zones. This message achieves communication across zones, making the platform accurately estimate the effect of the price difference.

TABLE I: Overall Performance Comparison

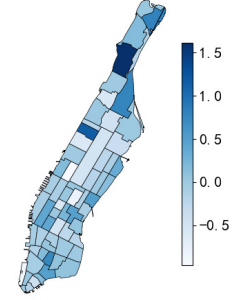
Metrics	Chengdu						Manhattan					
	ADR	ADR.Inc	OAR	OAR.Inc	OCR	OCR.Inc	ADR	ADR.Inc	OAR	OAR.Inc	OCR	OCR.Inc
COP	2,703,509.6	-	0.72	-	0.63	-	1,358,165.3	-	0.73	-	0.68	-
AP-RTRS	2,932,138.1	+8.46%	0.82	+13.89%	0.64	+1.59%	1,498,346.3	+10.32%	0.84	+15.07%	0.69	+1.48%
RSODR	3,252,671.5	+20.31%	0.81	+12.50%	0.66	+4.76%	1,522,290.1	+12.08%	0.85	+15.42%	0.71	+4.41%
MAP	3,528,299.1	+30.50%	0.80	+11.11%	0.70	+11.11%	1,632,721.7	+20.22%	0.81	+10.96%	0.74	+8.82%



(a) The Average Supply-demand Gap.



(b) Supply-demand Gap in Chengdu.



(c) Supply-demand Gap in Manhattan.

Fig. 4: Supply-demand Gap Performance.

Fig. 4(a) shows the supply-demand gap, i.e., the number of received orders subtracted from the number of received vehicles, in each zone within 60 seconds. The findings of the investigation unequivocally establish that MAP exhibits a heightened performance in maintaining supply-demand balance. Meanwhile, Fig. 4(b) and Fig. 4(c) show the average supply-demand gap in each zone of MAP, which proves that MAP ensures the supply-demand balance in most zones.

B. Impact of the Number of Zones

The ADR and OCR results for the four methods with varying numbers of zones in the Chengdu datasets are shown in Fig.5. As illustrated in Fig.5(a), for ADR, all four methods show an increase as the number of zones grows. A larger number of zones expands the action space available for crafting pricing strategies to maximize revenue, leading to higher ADR and OCR values. This happens because, on one hand, it enables more appropriate orders to share vehicles in fine-grained manner. On the other hand, it broadens the potential target zones for repositioning at a lower cost. Since MAP utilizes both ridesharing and repositioning, it consistently achieves the highest ADR among all methods. As for OCR in Fig. 5(b), the increase in zones lowers the minimum repositioning cost between zones, thereby alleviating the constraints on repositioning. The spatial messages exchanged between agents ensure that OCR continues to increase steadily.

C. Impact of Loss Function Weights

Table II presents the performance in terms of ADR and OCR with varying weighting factors w_m and w_a . For w_m , the mutual loss primarily focuses on capturing the influence of price differences across zones. A lack of mutual loss leads to faulty messages for coordination, worsening the performance. Conversely, when the mutual loss dominates the overall loss, the performances decline as the exaggerated messages ignore the spatial correlation across zones. For w_a , the attention mechanism becomes more crucial than the mutual representation itself, requiring a sufficiently large w_a to effectively filter messages.

D. Impact of Uncertain Behavior

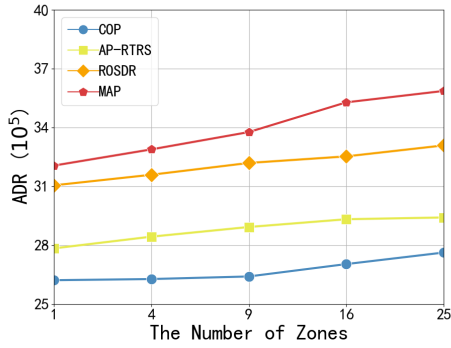
To explore the performance of the above methods in different order/vehicle behavior patterns, we further set two different cases, where the origin case is denoted by C1. In case C2, we reduce the probability of potential orders being converted to received orders to 80% of the probability in C1, i.e., $\xi_o^{C2}(a_t) = 0.8\xi_o(a_t)$, $\xi_o^{C2}(a_t) = 0.8\xi_o(a_t)$. In another case C3, we similarly reduce the probability of potential vehicles being converted to working vehicles to 80% of the value in C1. The simulation results for the three cases are presented in Fig. 6. MAP consistently achieves the highest ADR and OCR, demonstrating its robustness.

E. Ablation Study

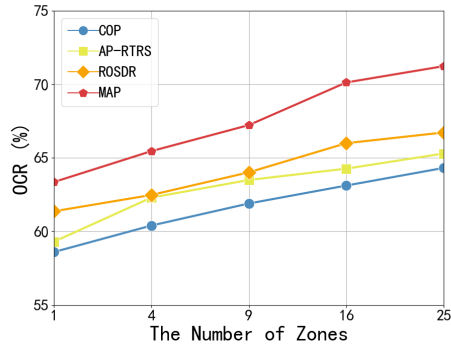
The effectiveness of mutual representation and attention is listed in Table. III. NMA is a basic version of the MAP

TABLE II: Comparison of ADR and OCR under different weight settings for mutual loss (w_m) and attention loss (w_a).

Value	Mutual Loss Weight (w_m)						Attention Loss Weight (w_a)					
	Chengdu			Manhattan			Chengdu			Manhattan		
	ADR	ADR Inc.	OCR	ADR	ADR Inc.	OCR	ADR	ADR Inc.	OCR	ADR	ADR Inc.	OCR
0	3,428,349.2	—	0.67	1,553,806.5	—	0.71	3,398,349.2	—	0.67	1,543,649.3	—	0.71
0.001	3,528,299.1	+2.92%	0.70	1,632,721.7	+5.08%	0.74	3,496,212.7	+2.88%	0.69	1,613,537.6	+4.53%	0.73
0.01	3,493,534.8	+1.90%	0.69	1,542,598.5	−0.72%	0.71	3,528,299.1	+3.82%	0.70	1,632,721.7	+5.77%	0.74
0.1	3,480,538.6	+1.52%	0.69	1,526,846.3	−1.74%	0.70	3,480,538.6	+2.42%	0.70	1,627,459.2	+5.43%	0.73



(a) ADR.



(b) OCR.

Fig. 5: The Impact of Number of Zones in Chengdu.

TABLE III: Ablation Study

Setting	Chengdu		Manhattan	
	ADR	OCR	ADR	OCR
NMA	3,263,175.4	0.66	1,532,372.4	0.70
NA	3,408,542.4 +4.45%	0.69	1,567,649.3 +2.30%	0.71
NM	3,421,306.6 +4.85%	0.69	1,619,586.2 +5.69%	0.73
MAP	3,528,299.1 +8.12%	0.70	1,632,721.7 +6.55%	0.74

strategy, which learns the policy by the RNN networks in agents. NM is a variant of MAP strategy, which aggregates the message $v_{nn'}$ based on z_t^n and n' without the mutual representation. NA is a variant of MAP strategy, which aggregates the message $v_{nn'}$ without attention-based weighted factor. The

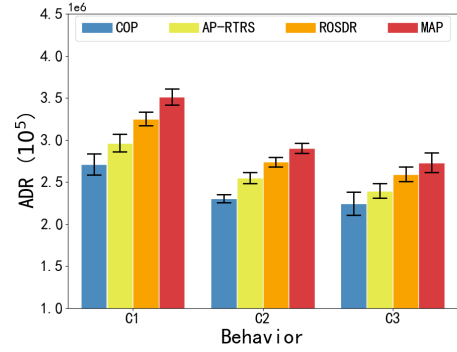


Fig. 6: The Impact of Uncertain Behavior.

gap between the performance of MAP and NA, NM, and NAM shows the efficiency of the attention mechanism and mutual representation. In Manhattan, the attention weighting factor achieved more performance improvement compared with that improved by mutual representation.

V. CONCLUSIONS

This study addressed the pricing problem by formulating it as an optimization problem, which was further transformed into a MARL framework aimed at maximizing overall revenue. To handle the coupled pricing strategies within the same zone and their influence caused by price differences across zones, we introduced MAP strategy based on mutual attention mechanism. First, MAP considered how price differences uniquely influenced coupled pricing strategies with ridesharing and repositioning. Second, we proposed an attention-based pricing message generation method based on mutual information theory. This approach accurately captured the mutual effects resulting from price differences. Lastly, we assessed the MAP framework's performance by measuring metrics such as the ADR, OCR, and other key indicators. Comprehensive simulation experiments were carried out to validate the effectiveness of the proposed approach.

VI. ACKNOWLEDGMENT

This work is sponsored in part by the National Science Fund for Distinguished Young Scholars (No. 62325208), in part by the National Natural Science Foundation of China (No. 62072330, 62272339), in part by the Beijing-Tianjin-Hebei Basic Research Cooperation Project (No. 24JCZXJC00050).

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